PHAM BUI QUANG MINH

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Link to my website portfolio: https://minhpahm09.github.io/PhamBuiQuangMinh/

EDUCATION

Amsterdam High School for the Gifted, Hanoi

GPA: Grade 9: 9.2/10.0 Grade 10: 9.4/10.0 Grade 11: 9.6/10.0
SAT: 1590 (Maths: 800 | Verbal: 790) **IELTS:** 8.0

Honors and Awards:

- Gold Award in the Asian Chemistry Olympiad, 2026 (International Level)
- Grand Prix in the Artificial Intelligence JAM, 2026 (International Level)
- Gold award in the World Invention Creativity Contest, 2026 (International Level)
- Bronze medal in the Indonesia International IoT Olympiad, 2026 (International Level)
- Bronze award in the Asian Chemistry Olympiad, 2025 (National Level)

RESEARCH

Co-author, "Research and Synthesis of $CoFe_2O_4$ Materials and Their Application in Drug Delivery".
Mentor: PhD Nguyen Cao Khang - Deputy Director of the Center for Nanotechnology and Science, Lecturer in the Faculty of Physics. Published in August, 2026.

Co-author, "Fabrication and Characterization of Copper Oxide Nanoparticles for Solar Cells".
Mentor: PhD Nguyen Cao Khang - Deputy Director of the Center for Nanotechnology and Science, Lecturer in the Faculty of Physics. Published in July, 2026.

CO-INVENTOR | PATENT March–October 2026
Title: *Process for Producing a Dual Nanocomplex of Silver Nanoparticles, Curcumin, and Berberine with Broad-Spectrum Antibacterial Efficacy*

- Developed a production process for a dual nanocomplex combining silver nanoparticles, curcumin, and berberine, designed to deliver broad-spectrum antibacterial activity.
- Received a patent granted by the Intellectual Property Office of Vietnam on October 9, 2026.

LEADERSHIP

Co-founder & Developer | Safedrive+ Feb 2026 – Present
Blood Alcohol Concentration monitoring model for safer driving decisions

- Researched ethanol absorption and BAC estimation, examining how alcohol dose, body weight, biological sex, and elapsed drinking time affect the concentration of ethanol in the bloodstream.
- Applied the Widmark Formula to convert alcohol consumption data into estimated BAC values, working with variables including alcohol mass, body weight, distribution factor, and time since drinking.
- Analyzed calculated BAC values against driving-safety thresholds to develop personalized recommendations, gaining experience in applying chemical concentration models to a real-world public-safety problem.

Co-founder | Dam Xanh May 2026 – Present

- Led a team of 16 to research and develop the "Healthy Flight Kit," designing regional meal concepts incorporating Vietnamese ingredients and cuisines with calorie and portion calculations for pilots, flight attendants, and active professionals around different dietary needs and work routines.
- Organized a public talk show on the gut microbiome and its relationship to diet and human health, featuring Prof. Dr. Nguyễn Trọng Hưng (National Institute of Nutrition) as speaker; coordinated program development and audience engagement for 40 participants.
- Allocated 80% of project proceeds to provide free meals for patients through the charity kitchen at Oncology Hospital.

Head of Laboratory & Experiments | Hóa ra dễ Oct 2025 – Jul 2026

- Directed 20 members in planning and executing chemistry demonstrations, overseeing task allocation, experiment preparation, and safety.
- Standardized 15 demonstrations for Chemis-mas Challenge, developing hands-on activities to explain core chemistry concepts.

- Led laboratory preparation for a Science Fair at Quoc Oai A Primary School in Hanoi, reviewing lesson plans and testing experiments to deliver safe, engaging science activities for 30 students.
- Coordinated science-fair preparation and demonstrations at Huu Lien Primary School in Lang Son, engaging 150+ elementary-school students through interactive experiments while supporting meal and essential-medicine donations to 60 disadvantaged children.

Head of Organizing Committee | SOS Hai Phong Village Oct 2025 – May 2026
Charity and science outreach program bringing hands-on chemistry activities to children

- Led a committee of 15 members to plan and deliver a charity program for 150+ children, delegating roles, setting the timeline, and coordinating with classmates, parents, and the homeroom teacher.
- Mobilized 13 million VND (~500 USD) in donations (school supplies, books, toys) and managed logistics, partnership with SOS Children's Village staff, and day-of execution, with the event running smoothly and on schedule.
- Designed and ran the "volcano eruption" chemistry demo and hosted as MC, directing minigames and the gift presentation.

Head of Logistics Department | OlympiAms Oct 2024 – May 2025
School-level selection competition for Hanoi–Amsterdam High School's "Road to Olympia" team

- Coordinated logistics for the October 2024 Demo Round and OlympiAms 2025 main competition, preparing 4 sets of competition equipment, stage materials, floral arrangements, and venue setups across 5 rounds.
- Managed on-site logistics for the 12 April 2025 Grand Final at Hall 200, coordinating 10 logistics members to arrange the competition stage, seating, check-in area, equipment, and audience setup for 160 attendees.
- Oversaw pre-event procurement, equipment checks, venue setup, and event-day breakdown, maintaining logistics readiness across 3 Semifinals, 1 Play-off, and 1 Grand Final.

WORK EXPERIENCE

Research Intern | Hanoi National University of Education May 2026 – Sep 2026

- Investigated $CoFe_2O_4$ magnetic nanoparticles for targeted drug delivery; synthesized and characterized 14–16 nm particles using XRD, TEM, and VSM.
- Analyzed magnetic properties, identifying room-temperature superparamagnetism and a saturation magnetization of ~55 emu/g; explored nanotoxicity and drug-delivery challenges.
- Synthesized CuO nanoparticles via chemical precipitation and characterized their crystal structure, optical absorption, and morphology using XRD, UV-Vis, and SEM.
- Evaluated CuO's potential as a solar-cell absorber by examining its nanoscale structure, semiconductor behavior, and light-absorption properties.
- Applied experimental design, chemical processing, material characterization, and data analysis skills to an independent project developing and testing a bio-based cleaning solution from pineapple peel waste.

OTHER ACTIVITIES

Chess Expertise | Ams Chess Academy Sep 2023 – Aug 2025

- Coordinated tournament preparation for the Hanoi Summer Chess Championship 2025, a city-wide competition that drew nearly 200 players, and the Hanoi Chess Tournament 2025, applying chess expertise to player coordination and competition operations.
- Organized chess sessions at Nguyễn Đình Chiểu School for 30+ students with disabilities, facilitating 15 simultaneous games and adapting game instructions and interaction to different student needs.

MC | VietSkill Sep 2026

- Received a certificate as an outstanding student from the VietSkill MC training course, organized and taught by editors from Vietnam Television (VTV).

SKILLS & INTERESTS

- **Skills:** Experimental Design, Chemical Synthesis, Nanomaterial Characterization, Process Optimization, Materials Analysis, XRD, UV-Vis, SEM, TEM, VSM, Data Analysis.
- **Interests:** Experimenting, Piano, Cooking, Chess.